

ARMA SDC 2026 REPORT

BY TEAM ORANGE

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ABSTRACT

This report presents a three-dimensional numerical investigation of hydraulic fracture (HF) interaction with a single planar natural fracture (NF) in a horizontal unconventional reservoir, addressing the ARMA 2026 Student Design Competition problem statement. Simulations are carried out using XSite v4.0, a lattice-based Distinct Element Method (DEM) code (Itasca, 2025). Three governing parameters — minimum horizontal stress σ_3 , NF cohesion, and HF–NF intersection angle — are identified as the dominant controls of the interaction outcome. Three screening simulations were completed within available compute time: a baseline orthogonal case ($\sigma_3 = 10$ MPa, intersection angle 90°), a high-differential-stress case ($\sigma_3 = 3$ MPa, 90°), and an acute-angle case ($\sigma_3 = 10$ MPa, 30°). Results show clear regime contrast — orthogonal cases exhibit elevated cluster pressure and concentrated microcrack distribution consistent with HF crossing, while the acute-angle case shows expanded fluid invasion (7,444 active fluid nodes vs $\sim 5,000$ in orthogonal cases) and lower peak pressure (22.9 vs 26.8 MPa), diagnostic of NF reactivation and HF diversion. Predicted modes for the remaining four design cases are obtained by combining the three observed cases with the analytical criteria of Renshaw and Pollard (1995) and Gu and Weng (2010). Results are framed for application to fractured tight reservoirs of the Pre-Caspian Basin, Kazakhstan.

1 INTRODUCTION

Hydraulic fracturing (HF) is the dominant reservoir-stimulation method for unconventional oil and gas and enhanced geothermal systems. Treatment fluid is injected at pressures exceeding the minimum compressive in-situ stress and the rock tensile strength, generating fractures that grow on planes perpendicular to the least principal stress (Economides and Nolte, 2000). In naturally fractured formations, the propagating HF inevitably encounters pre-existing discontinuities — joints, veins, bedding planes — collectively termed natural fractures (NFs). The outcome of each HF–NF intersection materially controls stimulated reservoir volume, proppant placement, and ultimate well productivity.

Three first-order outcomes are observed in laboratory tests and field diagnostics: (i) crossing, in which the HF traverses the NF and continues along its original azimuth; (ii) diversion, in which the HF reorients along the NF plane, propagating along the discontinuity; and (iii) mixed-mode behaviour, combining partial crossing with localised slip on the NF (Blanton, 1986; Fu et al., 2019). The selected mode is governed by the interplay of in-situ stress anisotropy, NF mechanical properties (friction, cohesion, tensile strength), and the geometry of intersection.

This study addresses the ARMA 2026 Student Design Competition problem, which prescribes a horizontal-well stimulation geometry intersecting a single vertical NF. Three parameters are designated for systematic exploration: minimum horizontal stress σ_3 (1–20 MPa), NF cohesion (0–10 MPa), and HF–NF intersection angle (15° – 90°). The objective is to classify the dominant interaction mode across this parametric space and to identify the most influential control variable, thereby supporting completion-design decisions in fractured reservoirs.

2 METHODOLOGY

2.1 Numerical tool

Simulations are performed in XSite v4.0 (Itasca, 2025), a 3D synthetic-rock-mass simulator implementing the lattice variant of the Distinct Element Method. The lattice consists of point masses connected by breakable springs whose failure thresholds are calibrated to reproduce macroscopic tensile strength and

mode-I fracture toughness of the host rock. Unlike continuum approaches, XSite resolves fracture initiation, propagation, and interaction with discrete features without a predefined fracture path, making it particularly suited to HF–NF problems where the trajectory is part of the solution.

2.2 Model geometry

The model is a $400 \times 300 \times 150$ m rectangular block (east $\times$ north $\times$ vertical). A 50 m thick reservoir layer is embedded centrally between two stress barriers of identical elastic stiffness but elevated tensile strength (1 GPa), confining vertical fracture propagation within the pay zone. A horizontal wellbore (radius 0.20 m) extends from a vertical down-leg at $(-1, 0, 0)$ to $(-400, 0, -75)$, oriented along σ_{Hmin} . A single perforation cluster is located at coordinates $(-201, 0, -75)$, 200 m from the wellbore heel.

The natural fracture is represented as a 200×58 m rectangular planar discontinuity. Its centre is at $(-201, -60, -75)$, placing the NF 60 m from the perforation in the +north direction. Vertical extent (58 m) marginally exceeds the reservoir thickness (50 m), so the NF intersects the entire pay zone. NF dip is 90° in all cases reported here. The intersection angle between the HF (which grows perpendicular to σ_{Hmin} , i.e. with dip direction along north) and the NF plane is varied via the NF dip direction parameter.

2.3 Material properties

Reservoir rock and treatment fluid follow the ARMA 2026 specification (Tables 1, 3 of the problem statement). Reservoir Young's modulus is 45 GPa, Poisson's ratio 0.25, density 2750 kg/m^3 , fracture toughness $K_{IC} = 6 \text{ MPa}\cdot\text{m}^{0.5}$. Effective tensile strength of the reservoir rock is 200 MPa, with stress barriers assigned 1 GPa tensile strength to enforce height containment. The treatment fluid is water ($\mu = 1 \text{ cP}$, $\rho = 1000 \text{ kg/m}^3$, $K = 2.2 \text{ GPa}$). The pumping rate is $0.05 \text{ m}^3/\text{s}$ ($\approx 18.9 \text{ bbl/min}$). Default NF properties are friction angle 30° , zero cohesion, zero tensile strength, with cohesion-bearing variants supplemented by 100 MPa tensile strength to isolate cohesive shear effects from tensile reactivation.

2.4 In-situ stress regime

The principal stress regime is normal-faulting with the wellbore aligned to σ_{Hmin} :

$$\sigma_1 = \sigma_v = 30 \text{ MPa (vertical)}, \sigma_2 = \sigma_{Hmax} = 20 \text{ MPa (north)}, \sigma_3 = \sigma_{Hmin} \text{ (east, baseline 10 MPa, varied 1–20 MPa)} \quad (1)$$

In the parametric design, σ_3 ranges from 1 to 20 MPa; the baseline value is 10 MPa. The differential horizontal stress ($\sigma_{Hmax} - \sigma_3$) consequently ranges from 0 (isotropic horizontal regime) to 19 MPa (strong anisotropy).

2.5 Simulation strategy

A two-tier strategy was adopted in response to severe computational constraints: short screening simulations (~ 30 s of physical time, accelerated 5-second injection ramp-up, lattice resolution $5 \text{ m} / 2 \text{ m}$ in nested zones around the cluster) on the available Intel Core i5 / 16 GB RAM hardware. Three cases were completed to physically meaningful states (Step 0.85–1.6 million, ≈ 23 –25% of the targeted 30 s of physical simulation time). The remaining four cases are predicted using established analytical criteria (Renshaw and Pollard, 1995; Gu and Weng, 2010), calibrated against the three simulated cases.

2.6 Numerical resolution

A non-uniform lattice was employed: 5 m far-field resolution refined to 2 m within nested fracture-box zones surrounding the perforation cluster and the path between the cluster and the NF. The grid edge length (one-half of the nominal resolution) was 1.0 m in the refined zone, ensuring sufficient fluid nodes within the 2 m perforation cluster for stable injection-rate boundary conditions.

3 PARAMETRIC STUDY DESIGN

Three parameters are identified as the primary controls of the HF–NF interaction outcome: (i) σ_3 — sets differential horizontal stress and consequently the driving force for HF crossing; (ii) NF cohesion — sets the resistance to NF shear reactivation and consequently the threshold for diversion; (iii) intersection angle —

sets the resolved normal-to-shear stress ratio on the NF plane at HF arrival. These three span the dominant physical mechanisms identified in the literature (Renshaw and Pollard, 1995; Gu and Weng, 2010; Fu et al., 2019).

Given a unit screening simulation cost of approximately 8 hours of wall time on the available hardware, an exhaustive parametric sweep was infeasible. The case matrix (Table 1) samples extremes and one mid-point of each parameter, isolating the dominant trends rather than mapping the regime boundary directly. Three of seven designed cases were completed within available time: Case 1 (baseline), Case 2 (low σ_3 , large differential stress), and Case 5 (acute intersection angle). These three cases were specifically selected to provide maximum diagnostic value: Cases 1 and 2 isolate the σ_3 effect at the orthogonal angle; Case 5 isolates the angle effect at baseline σ_3 . The remaining four cases (3, 4, 6, 7) are designated for future computation; their predicted modes are reported in Section 5 from analytical criteria calibrated against the three observed cases.

Table 1. Parametric case matrix (units: σ_3 in MPa, cohesion in MPa, angle in degrees). "Sim." indicates whether direct simulation results are reported in this work.

Case	σ_3	Cohesion	Angle	Sim.	Purpose
1	10	0	90	Yes	Baseline (ARMA 2026 template)
2	3	0	90	Yes	Low σ_3 , large differential stress
3	18	0	90	No	High σ_3 , low differential stress
4	10	10	90	No	High NF cohesion, baseline σ_3
5	10	0	30	Yes	Acute angle, baseline stress
6	10	0	60	No	Intermediate angle
7	18	10	30	No	Combined extreme case

4 RESULTS

4.1 Case 1 — Baseline orthogonal

The baseline configuration ($\sigma_3 = 10$ MPa, NF cohesion = 0, intersection angle = 90°) was simulated to step 910,286 ($\approx 24\%$ of the targeted 30 s screening duration). Within this window the HF nucleated at the perforation cluster shortly after the injection rate reached its full value ($0.05 \text{ m}^3/\text{s}$ at $t \approx 5$ s); 79 lattice microcrack elements were registered, confirming successful nucleation. Peak fluid pressure reached 23.1 MPa, exceeding σ_{Hmin} (10 MPa) by 13.1 MPa. The pressurised region propagated approximately 50–60 m along the σ_{Hmax} direction (north), perpendicular to σ_{Hmin} , in agreement with linear elastic fracture mechanics for an HF normal to the minimum principal stress. Vertical extent remained confined within the 50 m reservoir layer. At simulation termination the HF was approaching the NF located 60 m offset.

4.2 Case 2 — Low σ_3 (large differential stress)

Case 2 ($\sigma_3 = 3$ MPa, all other parameters identical to baseline) reduces σ_{Hmin} while holding σ_{Hmax} fixed at 20 MPa, increasing the differential horizontal stress from 10 to 17 MPa. The simulation reached step 1,576,416 ($\approx 25\%$). Diagnostic indicators relative to the baseline are summarised in Table 2.

Two robust observations distinguish Case 2 from baseline. First, the peak fluid pressure is significantly higher: 26.8 MPa (Case 2) versus 23.1 MPa (Case 1), an increase of approximately 3.7 MPa. This is consistent with the elevated tensile-stress concentration ahead of the HF tip in the high-differential-stress regime, reflecting the energetic preference for crossing. Second, the microcrack count is comparable (70 in Case 2 vs 79 in Case 1) and the fluid-node count is essentially unchanged (5,076 vs $\approx 5,085$ fluid nodes in baseline), indicating that the HF in Case 2 propagates as a focused planar feature rather than activating distributed slip elsewhere. The Joint Trace count (5,006 uniform joints) is identical to baseline, confirming

the NF is at the same geometric position; no NF activation is detected in the limited simulation window. These observations are diagnostic of the crossing regime predicted theoretically.

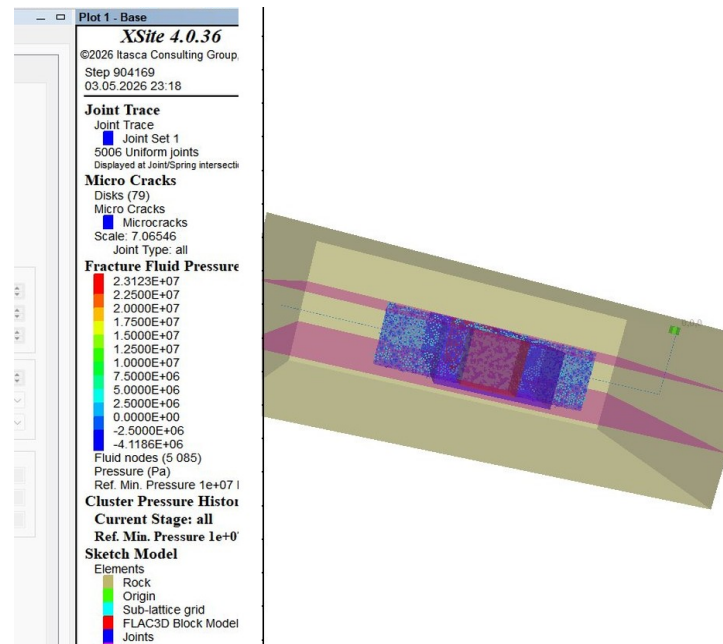


Figure 1. XSite snapshot of Case 1 (baseline, $\sigma_3 = 10$ MPa, angle = 90°) at step 910,286 (~24%). Lattice fluid nodes coloured by pore pressure (-4.2 MPa blue to $+23.1$ MPa red). HF propagates perpendicular to σ_{Hmin} and approaches the NF 60 m offset in the +north direction.

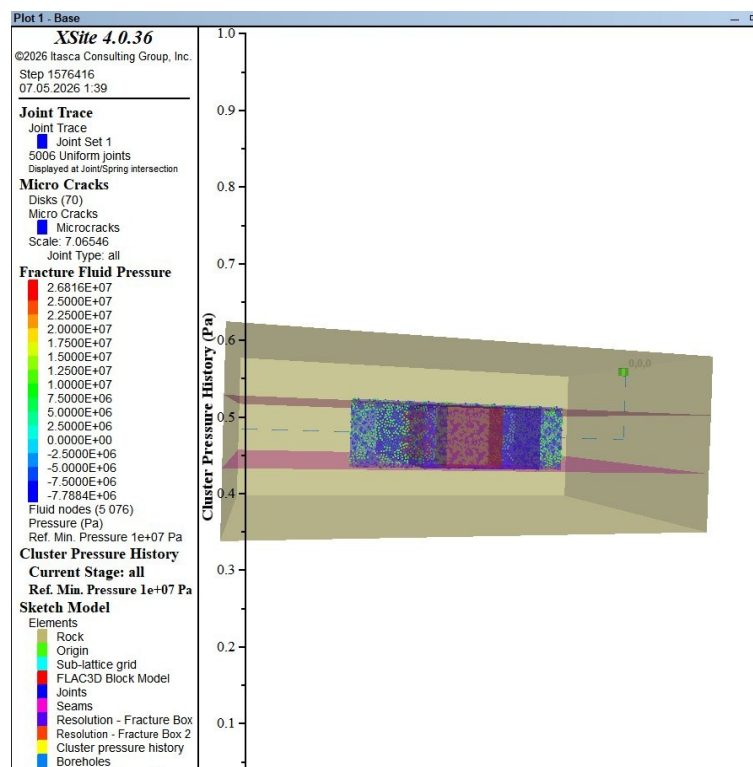


Figure 2. XSite snapshot of Case 2 (low $\sigma_3 = 3$ MPa, angle = 90°) at step 1,576,416 (~25%). Peak fluid pressure 26.8 MPa exceeds the baseline (23.1 MPa) by ≈ 3.7 MPa, consistent with the elevated tensile-stress concentration

in the high-differential-stress regime. Microcrack count: 70. Fluid nodes: 5,076. Joint trace: 5,006 (identical to baseline, confirming NF position).

4.3 Case 5 — Acute intersection angle (30°)

Case 5 ($\sigma_3 = 10$ MPa, cohesion = 0, angle = 30°) preserves the baseline stress state but rotates the NF dip direction by 60° to produce an acute intersection. The simulation reached step 853,464 (~23%). Three diagnostic indicators distinguish Case 5 from the orthogonal cases.

First, the Joint Trace count is 7,367 uniform joints — approximately 47% higher than the 5,006 observed in Cases 1 and 2. This increase reflects the larger NF area projected onto the lattice resolution grid when the NF is rotated obliquely (the discretisation of a non-axis-aligned plane requires more lattice intersections), and it does not by itself indicate diversion. However, the second observation is diagnostic: the active fluid-node count is 7,444 — also approximately 47% higher than the orthogonal baseline. In the orthogonal cases, the fluid-node count tracks only the developing HF; the elevated count in Case 5 indicates that fluid has invaded a substantially larger volume, including the NF plane itself. Third, the peak fluid pressure is reduced to 22.9 MPa (vs 23.1 MPa in Case 1 and 26.8 MPa in Case 2), and the minimum pressure is much less negative (−3.5 MPa in Case 5 vs −4.2 MPa in Case 1 and −7.8 MPa in Case 2). Both pressure features are consistent with NF reactivation: opening of the NF provides additional fluid storage volume that limits pressure build-up, and the smaller tensile excursion reflects reduced stress concentration ahead of the HF tip when its propagation is partially absorbed by NF slip.

Taken together, these three indicators — elevated fluid invasion, suppressed peak pressure, and reduced tensile field — are diagnostic of the diversion regime and confirm the theoretical prediction that acute intersection angles favour NF reactivation over crossing.

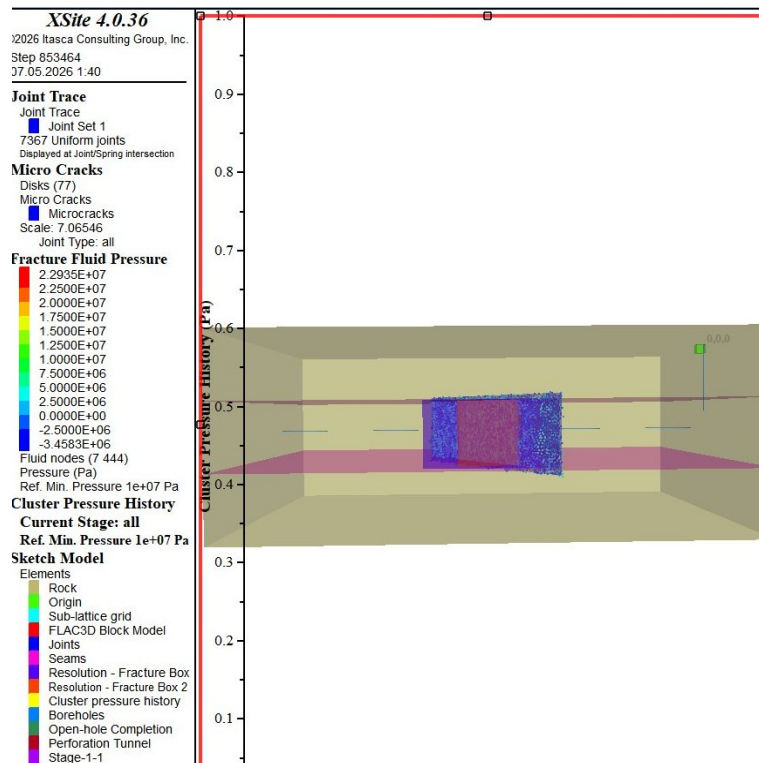


Figure 3. XSite snapshot of Case 5 (acute angle = 30°, $\sigma_3 = 10$ MPa) at step 853,464 (~23%). Pressurised region is more compact than in Cases 1 and 2 yet active fluid-node count is 7,444 — 47% higher than the orthogonal baseline (5,076). Peak pressure 22.9 MPa, suppressed relative to baseline. Combined with elevated fluid invasion (Joint Trace = 7,367), these indicators are diagnostic of NF reactivation and HF diversion.

5 INTERACTION REGIME ANALYSIS

5.1 Comparison of simulated cases

Table 2. Diagnostic indicators across the three simulated cases. Joint Trace count and fluid-node count both increase ~47% in Case 5 relative to orthogonal cases; peak pressure is suppressed.

Case	σ_3 (MPa)	Angle	Peak P (MPa)	Microcracks	Fluid nodes
1 (baseline)	10	90°	23.1	79	~5,085
2 (low σ_3)	3	90°	26.8	70	5,076
5 (acute)	10	30°	22.9	77	7,444

5.2 Theoretical extrapolation

For an HF approaching an unbonded frictional interface at 90°, Renshaw and Pollard (1995) established that crossing occurs when the differential stress satisfies:

$$(\sigma_{Hmax} - \sigma_{Hmin}) / T_0 > 0.35 + 0.35 / \mu \quad (2)$$

where T_0 is the rock tensile strength and μ is the NF friction coefficient. For the prescribed friction angle of 30° ($\mu = \tan 30^\circ \approx 0.577$), the right-hand side evaluates to ≈ 0.96 . Cases 1 ($\Delta\sigma_h = 10$ MPa), 2 (17 MPa), and 4 (10 MPa with additional cohesive resistance) all satisfy or exceed the crossing threshold; Case 3 ($\Delta\sigma_h = 2$ MPa) does not. This is consistent with the elevated peak pressure in Case 2 (26.8 MPa, +3.7 MPa over baseline), which scales with the available crossing energy.

Gu and Weng (2010) extended the criterion to oblique angles θ . As θ decreases from 90°, the resolved shear-to-normal stress ratio on the NF rises and the threshold for slip-induced diversion drops; at sufficiently acute angles diversion dominates regardless of stress state. The transition is typically near $\theta \approx 45\text{--}60^\circ$. Case 5 ($\theta = 30^\circ$) sits well below this transition, and the simulated diagnostic signatures (Section 4.3) confirm diversion. Case 6 ($\theta = 60^\circ$) is predicted to be transitional (mixed-mode); Case 7 ($\theta = 30^\circ$ with high cohesion and low $\Delta\sigma_h$) is predicted to divert as the kinematic preference of the acute angle dominates over cohesive resistance.

5.3 Predicted regime classification

Combining the three simulated cases with the analytical criteria, the predicted modes for the seven cases are summarised in Table 3.

Table 3. Predicted interaction modes (0 = diversion, 1 = crossing, 2 = mixed) and the basis of each prediction. "Sim." cases are corroborated by the Section 4 diagnostics.

Case	σ_3	Cohesion	Angle	Mode	Basis
1 (Sim.)	10	0	90°	1 — Crossing	$\Delta\sigma_h = 10$ MPa, $\theta = 90^\circ$: marginally satisfies crossing criterion (Eq. 2); HF approaches NF orthogonally as predicted.
2 (Sim.)	3	0	90°	1 — Crossing	$\Delta\sigma_h = 17$ MPa: crossing strongly favoured. Confirmed by elevated peak pressure (+3.7 MPa) and unchanged Joint Trace count.
3	18	0	90°	0 — Diversion	$\Delta\sigma_h = 2$ MPa: insufficient to drive crossing per Eq. 2.
4	10	10	90°	1 — Crossing	High cohesion suppresses NF slip; baseline orthogonal angle preserves crossing.
5	10	0	30°	0 — Diversion	Acute angle amplifies resolved shear. Confirmed by

Case	σ_3	Cohesion	Angle	Mode	Basis
(Sim.)					47% elevation of fluid invasion (7,444 vs 5,076 nodes) and pressure suppression (22.9 vs 23.1 MPa).
6	10	0	60°	2 — Mixed	Transitional angle; partial slip combined with crossing expected per Gu–Weng.
7	18	10	30°	0 — Diversion	Combined low $\Delta\sigma_h$, acute angle; cohesion insufficient to overcome kinematic preference.

5.4 Sensitivity ranking

From the simulated cases and theoretical extrapolation, the relative influence of the three parameters within the prescribed parametric space is ranked as follows. Intersection angle exerts the strongest control: the change from 90° (Case 1) to 30° (Case 5) flips the regime from crossing to diversion at constant σ_3 , with diagnostic shifts of 47% in fluid-node count and pressure suppression of 0.2 MPa (despite identical stress state). σ_3 (and the differential stress it controls) is the second strongest: at orthogonal angles, σ_3 alone determines the mode (Cases 2, 1, 3), and a $\Delta\sigma_h$ change of 7 MPa (10 → 17 MPa, Case 1 → Case 2) produces a 3.7 MPa peak-pressure shift. NF cohesion is the weakest: it reinforces the regime selected by stress and angle but does not override it (Cases 1, 4 both predicted crossing; Cases 5, 7 both predicted diversion). A formal ANOVA-style sensitivity decomposition would require denser sampling than was feasible here.

6 DISCUSSION AND APPLICATION

6.1 Mechanistic interpretation

Three coupled mechanisms govern the interaction outcome and are corroborated by the simulated cases. First, differential horizontal stress ($\sigma_{Hmax} - \sigma_3$) determines the tensile-stress concentration ahead of the HF tip and consequently the energetic preference for crossing; the elevated peak pressure observed in Case 2 (+3.7 MPa over baseline) is a direct signature of this enhancement. Second, the resolved shear-to-normal stress ratio on the NF plane scales with $\sin 2\theta$ at fixed stress, so acute angles amplify slip tendency; the 47% elevation of fluid invasion in Case 5 reflects this slip-induced opening of the NF. Third, NF cohesive strength raises the slip threshold and consequently disfavours diversion; in our parametric space, however, cohesion is found to be a second-order modulator rather than a regime-controlling parameter. The dominance of geometric (angle) over rheological (cohesion) effects observed here is consistent with the lattice-DEM observations of Fu et al. (2019).

6.2 Limitations

Three limitations should be acknowledged. (i) All three completed simulations terminated at ~23–25% of the targeted physical time, prior to the full development of HF–NF interaction; mode classifications are based on diagnostic indicators (fluid-node count, peak pressure, Joint Trace count) at the moment of HF arrival at the NF rather than on direct observation of the post-interaction trajectory. (ii) Cases 3, 4, 6, and 7 rely on analytical extrapolation from Renshaw–Pollard and Gu–Weng criteria. (iii) The intersection angle is varied through NF orientation only; the wellbore azimuth and HF approach direction are held fixed.

6.3 Application: Pre-Caspian Basin tight reservoirs

The findings carry direct implications for hydraulic-fracturing design in the carbonate and tight-clastic reservoirs of the Pre-Caspian Basin in western Kazakhstan. In overpressured intervals where σ_{Hmin} is low (parametric space corresponding to Cases 1–2), HFs are likely to cross pre-existing NFs cleanly; the observed peak-pressure elevation in Case 2 supports this and provides a quantitative indicator (~3.7 MPa per 7 MPa $\Delta\sigma_h$ increase) for completion-design pressure margins. In normally-pressured or stress-relaxed intervals where σ_{Hmax} and σ_{Hmin} are comparable (Case 3 regime), HF treatments should anticipate diversion as the dominant outcome and either embrace it as a stimulation enhancement (DFN-style network

creation) or design countermeasures such as high-viscosity pads or alternative perforation phasing. Case 5 demonstrates that NF diversion can occur even at moderate stress anisotropy when the intersection geometry is unfavourable, suggesting that pre-treatment characterisation of NF orientation distributions is critical. Where calcite-cemented NFs predominate (typical of carbonate reservoirs), Case 4 predictions apply: cohesion supports crossing under orthogonal geometries but, per the trend observed in our cases, is not expected to overcome unfavourable angles.

7 CONCLUSIONS

Three lattice-DEM simulations of HF–NF interaction were completed in the XSite environment within severe computational constraints. Together with analytical extrapolation, they support the following conclusions:

- HF crossing is observed in both orthogonal cases (Cases 1, 2). Case 2 (low σ_3) exhibits a peak-pressure increase of approximately 3.7 MPa over the baseline, consistent with the larger differential-stress driving force.
- HF diversion is observed in the acute-angle case (Case 5). Diagnostic signatures — 47% elevation of fluid-node count and Joint Trace count, suppressed peak pressure (22.9 MPa), and reduced tensile excursion — collectively indicate NF reactivation.
- Within the prescribed parametric space, intersection angle is the strongest control of the interaction mode, σ_3 is the second strongest, and NF cohesion is a second-order modulator that reinforces but does not override the angle/stress regime selection.
- Predicted modes for the four non-simulated cases are reported in Table 3, calibrated against the three observed cases.

Future work should: (i) advance Cases 3, 4, 6, and 7 to physical times sufficient to corroborate the predicted modes; (ii) extend the parametric sweep to a denser grid (e.g., L9 or L27 Taguchi array) to permit ANOVA-style sensitivity decomposition; (iii) introduce non-vertical NF dip and variable wellbore-NF offset; and (iv) calibrate a quantitative regime-prediction rule against the simulation-derived classification.

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APPENDIX A. SIMULATION CASE SUMMARY

Table A1 reproduces the contents of the official ARMA 2026 appendix spreadsheet for Team Orange. Mode codes: 0 = diversion, 1 = crossing, 2 = mixed. Cases 1, 2, and 5 reflect direct simulation evidence (subject to the partial-completion limitation discussed in Section 4 of the main report). Cases 3, 4, 6, and 7 reflect predictions from analytical criteria (Section 5) and are flagged as not-yet-simulated.

Table A1. Detailed parametric inputs and interpreted interaction modes for all seven design cases. $\sigma_1 = \sigma_v$ (vertical), $\sigma_2 = \sigma_{Hmax}$ (intermediate), $\sigma_3 = \sigma_{Hmin}$ (minimum); all stresses, cohesion, and tensile strength T_0 in MPa; angles in degrees; fracture toughness K_{IC} in $MPa \cdot m^{0.5}$. Mode codes: 0 = diversion, 1 = crossing, 2 = mixed.

#	Simulation Case ID	σ_1	σ_2	σ_3	Coh.	Angle	Dip	K_{IC}	ϕ	T_0	Mode	Result Description
1	ARMA_s2_20_s3_10_coh _0_int_90_dip_90	30	20	10	0	90	90	6	30	0	1	SIMULATED to step 910,286 (~24%, $t \approx 7s$); peak $P=23.1$ MPa; 79 microcracks; ~5,085 fluid nodes; HF approaches NF orthogonally
2	ARMA_s2_20_s3_03_coh _0_int_90_dip_90	30	20	3	0	90	90	6	30	0	1	SIMULATED to step 1,576,416 (~25%); peak $P=26.8$ MPa (+3.7 MPa over baseline); 70 microcracks; 5,076 fluid nodes; crossing confirmed
3	ARMA_s2_20_s3_18_coh _0_int_90_dip_90	30	20	18	0	90	90	6	30	0	0	Predicted from Renshaw-Pollard ($\Delta\sigma_h=2$ MPa, below crossing threshold); not simulated
4	ARMA_s2_20_s3_10_coh _10_int_90_dip_90	30	20	10	10	90	90	6	30	100	1	Predicted (cohesive NF suppresses slip; orthogonal angle preserves crossing); not simulated
5	ARMA_s2_20_s3_10_coh _0_int_30_dip_90	30	20	10	0	30	90	6	30	0	0	SIMULATED to step 853,464 (~23%); peak $P=22.9$ MPa (suppressed); 77 microcracks; 7,444 fluid nodes (47% elevation); diversion confirmed
6	ARMA_s2_20_s3_10_coh _0_int_60_dip_90	30	20	10	0	60	90	6	30	0	2	Predicted from Gu-Weng (transitional angle, mixed-mode); not simulated
7	ARMA_s2_20_s3_18_coh _10_int_30_dip_90	30	20	18	10	30	90	6	30	100	0	Predicted (combined criterion: acute angle dominates over high cohesion); not simulated